

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

In the xy -plane, line k has a slope of 5 and a y -intercept of $(0, -35)$. What is the x -coordinate of the x -intercept of line k ?

Correct Answer: 7

Rationale

The correct answer is 7 . An equation of a line in the xy -plane can be written in the form $y = mx + b$, where m is the slope of the line and $(0, b)$ is the y -intercept of the line. It's given that line k has a slope of 5 and a y -intercept of $(0, -35)$. Therefore, $m = 5$ and $b = -35$. Substituting 5 for m and -35 for b in the equation $y = mx + b$ yields $y = 5x - 35$. The x -intercept of a line in the xy -plane is the point where the line intersects the x -axis, which is a point with a y -coordinate of 0 . Substituting 0 for y in the equation $y = 5x - 35$ yields $0 = 5x - 35$. Adding 35 to both sides of this equation yields $35 = 5x$. Dividing both sides of this equation by 5 yields $7 = x$. Therefore, the x -coordinate of the x -intercept of line k is 7 .